

Day 5: Network Testing Tools Theory :Tools for network testing: Ping, Traceroute, nslookup. Practical :Use these tools to test connectivity in a LAN. Task: Report on the tools used and results obtained.

Tasks :-

1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Ans.:-

```
C:\Users\Dell>ping -4 www.youtube.com

Pinging www.youtube.com [142.251.156.4] with 32 bytes of data:
Reply from 142.251.156.4: bytes=32 time=10ms TTL=116
Reply from 142.251.156.4: bytes=32 time=10ms TTL=116
Reply from 142.251.156.4: bytes=32 time=10ms TTL=116
Reply from 142.251.156.4: bytes=32 time=10ms TTL=116

Ping statistics for 142.251.156.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 10ms, Average = 10ms
```

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

Hint: Each hop represents a network device your data passes through to reach the destination.

Ans.:-

```
C:\Users\Dell>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.64.140.250]
over a maximum of 30 hops:

  0  1 ms    1 ms    1 ms  gpon.net [192.168.1.1]
  1  3 ms    3 ms    3 ms  10.100.0.1 [10.100.0.1]
  2  *        *        4 ms  10.233.11.2 [10.233.11.2]
  3  7 ms    6 ms    7 ms  115.242.212.97
  4  6 ms    7 ms    6 ms  172.31.3.218 [172.31.3.218]
  5  6 ms    7 ms    6 ms  192.168.188.50 [192.168.188.50]
  6  7 ms    7 ms    7 ms  a23-64-140-129.deploy.static.akamaitechnologies.com [23.64.140.129]
  7  6 ms    7 ms    7 ms  a23-64-140-250.deploy.static.akamaitechnologies.com [23.64.140.250]

Trace complete.
```

3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Ans.:-

```
C:\Users\Dell>nslookup www.zomato.com
Server:  gpon.net
Address:  fe80::1

Non-authoritative answer:
Name:    e70929.dsca.akamaiedge.net
Addresses:  2600:140f:1e00::1737:f4d0
            2600:140f:1e00::1737:f4c3
            49.44.140.145
            49.44.140.187
Aliases:  www.zomato.com
            zomato.edgekey.net
```

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Ans.:-

```
C:\Users\Dell>ping 192.168.1.8

Pinging 192.168.1.8 with 32 bytes of data:
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128
Reply from 192.168.1.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

If the device is switched off, disconnected from Wi-Fi, or not connected to the network, the ping request does not receive a reply. Instead, the Command Prompt displays messages such as:

Request timed out. Or Destination host unreachable.

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

Constraint: Give a specific example for each tool based on your own test results above.

Ans.:-

Network Diagnostic Tools Report:-

To troubleshoot different types of network problems, I used the ping, tracert (traceroute), and nslookup commands.

The ping command is most useful for checking whether a website or device is reachable. During my test, ping www.youtube.com returned "Request timed out", indicating that the destination did not respond to the ping request, so connectivity or ICMP blocking could be the cause.

The tracert command is the best tool for diagnosing a slow internet connection because it displays every intermediate router (hop) between my computer and the destination. By tracing the route to www.flipkart.com, I could identify where delays might occur if one of the hops had a high response time.

The nslookup command is ideal for troubleshooting DNS-related problems because it checks whether a domain name is being translated into an IP address correctly. In my test, nslookup www.zomato.com successfully returned the website's IP address, confirming that DNS resolution was working.

These three tools complement each other: ping verifies basic connectivity, tracert identifies routing and latency issues, and nslookup confirms DNS functionality. Using them together provides a systematic way to diagnose common network problems such as websites not loading, slow connections, and domain name resolution failures.